Dalfin Rafael Solihin



Mechatronics Student · AI Automation, LLM Pipelines & Embedded Systems

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PROFILE

Mechatronics student at HM Munich seeking a Werkstudent role in automation and AI tooling — Python and API integration, LLM and RAG pipelines, low-code workflows — turning manual, repetitive work into something that runs itself. Fluent German (C1) and English, with a year of customer-facing work in Munich.

PROJECTS

Financial Report Q&A System (RAG Pipeline)

Python, LangChain, ChromaDB, Gemini

- Built a retrieval-augmented generation system answering natural-language questions over **DAX annual reports** (Siemens, BMW), grounding every answer in the filing text rather than model memory
- Engineered the **pdfplumber** ingestion pipeline: per-page extraction, metadata parsing, OCR detection, and per-file error isolation so one corrupt PDF cannot halt a batch
- Implemented sentence-transformer embeddings, ChromaDB storage and a LangChain RetrievalQA chain, plus a strict-JSON extraction mode — ratios (ROE, margin, debt-to-equity) computed in **plain Python rather than trusting LLM arithmetic**

AI Application Generator (CV Automator)

Python, Gemini API, Streamlit

- Local, privacy-first tool generating tailored CVs and cover letters from an Obsidian vault: prompt engineering maps unstructured notes to job requirements, and a human-in-the-loop pipeline strips personally identifiable information before any API call, re-injecting it during local PDF assembly. Built with agentic coding tooling including **Claude Code**.
- **Autonomous Arduino Mop Bot** (Arduino C++, Onshape CAD, 3D printing) — designed and 3D-printed the chassis, programmed autonomous navigation with ultrasonic obstacle detection, integrated microcontroller, motor drivers, sensors and power into one working system
- **STM32 State Machine & Timer Controller** (C, hardware timers) — timer-driven finite state machine for multi-function button input, with a software debouncing module and non-blocking state transitions
- **YOLOv8 object classifier** (Python, Ultralytics) — trained a custom detection model on a self-prepared dataset to distinguish visually similar classes

EXPERIENCE

Student Research Assistant

Apr 2026 – Jul 2026

Munich University of Applied Sciences (HM) — Munich · 10h/week

- Automated analysis of SFRA measurement data with a custom **Python** script, replacing manual evaluation so measurements could be re-run the same day
- Conducted Sweep Frequency Response Analysis on rotating electrical machines to detect short-circuits and earth faults; built the circuits and data-acquisition setups
- Worked with researchers and doctoral students to optimize testing procedures

Retail Sales Assistant (Verkäufer) — Vinzenz Zöttl Bäckerei, Munich

Sep 2025 – Jan 2026

- Advised customers and handled POS and cash operations in German in a high-traffic retail environment, to HACCP hygiene and quality standards

Hospitality Service & Event Assistant — Weehive GmbH, Munich

Feb 2025 – Oct 2025

- Banquet and guest service at Hotel Bayerischer Hof, BMW Welt and the Bauma 2025 trade fair — fast-paced teamwork and cross-cultural communication under time pressure

EDUCATION

B.Eng. Mechatronics (in progress)

Oct 2024 – present

Munich University of Applied Sciences (HM) — Munich, Germany

Feststellungsprüfung T-Kurs (Technik) — Grade 1.5

Mar 2024 – Jul 2024

Internationales Studienkolleg, Hochschule Kaiserslautern — Germany

- Abitur-equivalent university-entrance qualification, examined in German — Physik 1.0, Mathematik/Informatik 1.7, Deutsch 2.0

TECHNICAL SKILLS

AI & Automation	LangChain, HuggingFace (sentence-transformers), RAG pipelines, ChromaDB, Google Gemini API, prompt engineering, Claude Code, REST API integration, OCI
Programming	Python, C/C++ (STM32 HAL, FSM, timer logic, debounce), VHDL, Verilog, MATLAB
Platforms	STM32 (ARM Cortex-M), ESP32, Arduino
Tools	Git, GitHub, STM32CubeIDE, VSCode, Streamlit, KiCad, CATIA V5, Onshape, 3D printing, MS Office
Spoken	Indonesian (native) · English (fluent, C2) · German (fluent, C1) · Chinese (advanced, B2)
Activities	AkaModell München, aerodynamics (2026–) · Event MC & Stage Manager (2025)